

Assessing Ranking Ability in Low Default Portfolio Models

A Tendency-Based Metric Framework

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Low Default Portfolios

- Qualitative definitions of Low Default Portfolios (LDPs) leave considerable room for interpretation, as practitioners may disagree on whether a particular portfolio should be classified as a low default portfolio.
- In practice, when assessing whether a portfolio qualifies as an LDP, practitioners typically consider both the level of default rates and the number of observed defaults over the available historical period.
- However, the defining characteristic of an LDP is the limited number of observed defaults, which reduces the reliability and statistical validity of risk parameter estimates based on historical data.
- Because of the limited number of observed defaults, LDP models are often developed using expert judgment rather than relying solely on statistical methods.
- On one hand, the combination of limited default data and expert judgment creates significant challenges for the development and evaluation of Probability of Default (PD) models. On the other hand, robust validation of LDP models is particularly important, as these portfolios often represent a significant share of a bank's overall exposure.

Low Default Portfolios cont.

- Modeling PD for LDPs is inherently challenging because the small number of defaults limits the statistical reliability of estimates from historical default data.
- As a result, PD risk differentiation functions for LDPs are typically developed by combining expert judgment and statistical methods, primarily to identify the most relevant risk factors.
- Model evaluation and validation are equally challenging, as many statistical tests developed for high-default portfolios have limited applicability to LDPs.
- Consequently, qualitative assessment and benchmarking against external ratings have become common validation approaches for LDPs. However, benchmarking requires the availability of external ratings and relies on assumptions that may not hold in all cases.
- In practice, model evaluation and validation of LDPs are often more focused on calibration than on assessing the model's ranking ability.
- To support the development of new approaches for assessing the ranking ability of LDP models, this presentation proposes a tendency-based approach. To the best of the author's knowledge, this method has not yet been used in practice.

Tendency-Based Approach to Model's Ranking Ability Testing

- Current practice for assessing ranking ability in LDPs typically relies on qualitative, case-by-case reviews of individual defaulters by examining how their ratings change as they approach default.
- This approach lacks a formal and quantifiable metric, making it difficult to draw consistent conclusions about a model's ranking ability. Moreover, it does not compare changes in defaulters' ratings with those of the performing population. As a result, the assessment is prone to subjective interpretation and is difficult to reproduce or defend during audits or regulatory reviews.
- To address some of these limitations, [Andrija Djurovic](#) proposes a new tendency-based approach for assessing the ranking ability of LDP models.
- The proposed framework decomposes ranking ability into a static and a dynamic dimension and introduces three tendency-based metrics.

Tendency-Based Approach to Model's Ranking Ability Testing cont.

- The static dimension addresses the question: Are defaulters, on average, assigned worse ratings than performing obligors?
- The dynamic dimension addresses two questions: Does the gap between defaulters and performers widen as default approaches? and Do defaulters' ratings deteriorate as they approach default?
- The static metric is defined by the average residual, which measures the average difference between defaulters' ratings and the baseline rating of performing obligors.
- The dynamic dimension is described by two metrics: the residual trend and the rating trend. The residual trend measures whether the gap between the average ratings of defaulters and performers widens as default approaches. Unlike the residual trend, the rating trend is an absolute measure that evaluates whether defaulters' ratings deteriorate as the obligor approaches default.
- As can be seen from the definition of the proposed metrics, the number of reference periods before default must be specified before applying the framework.
- Furthermore, the analysis is performed relative to the default date of each defaulter, so preparing the data requires particular attention.

Tendency-Based Approach to Model's Ranking Ability Testing cont.

- In general, the expected value of each proposed metric is greater than zero. However, different combinations of metric values are possible, and the overall assessment of the model's ranking ability should consider both their magnitudes and their expected distributions.
- Each metric can be formally tested against zero using a bootstrap approach. For reliable results, a minimum of five observed defaults is recommended.
- Practitioners are encouraged to further explore the impact of the minimum number of defaults, the choice of reference periods before default, and possible extensions of the framework based on quasi-default obligors.
- The following slides present a simulation study illustrating the calculation of each proposed metric and the estimation of their distributions using a bootstrap-based approach.

Simulation Study

The purpose of this simulation is to illustrate concrete values for the calculation of each proposed metric, along with their bootstrap-based distributions. The simulation relies on the following assumptions:

- 1 a seven-grade rating scale, where 1 represents the best rating and 7 the worst;
- 2 five observed default events;
- 3 1,000 performing obligors identified relative to the default events;
- 4 eight quarters prior to default are analyzed.

Given these assumptions and the simulated dataset available at this [link](#), the following slides present samples of the performing portfolio and defaulters' data, the main results of the tendency-based metric calculations, as well as their averages and bootstrap-based distributions.

Due to the limited number of observations, bootstrapping is performed over the full set of default combinations $5^5 = 3,125$.

Note:

The presented examples and assumptions are for illustrative purposes only; practitioners are encouraged to adapt the proposed framework to their specific use cases.

Simulation Study cont.

The following tables present a sample of performing and defaulted obligors by reference quarter relative to the eight quarters preceding the individual default date:

#performing portfolio

##		Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
##	Performing_1	5	4	4	4	4	4	4	5
##	Performing_2	3	3	2	2	2	2	2	2
##	Performing_3	4	4	4	4	4	4	4	4
##	Performing_4	4	4	4	4	4	4	4	4
##	Performing_5	3	4	4	4	4	3	2	1
##	Performing_6	1	1	1	1	1	1	1	1

#defaulted portfolio

##		Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
##	Default_1	4	4	5	5	5	5	6	7
##	Default_2	5	6	6	7	7	7	7	7
##	Default_3	4	4	4	5	6	6	7	7
##	Default_4	5	5	6	6	6	7	7	7
##	Default_5	4	5	5	7	7	7	7	7

Simulation Results

The following table presents the baseline ratings of the performing portfolio. As defined in the framework, baseline ratings are used to calculate the average residual and the residual trend.

##	quarter	baseline
## 1	Q1	4.013
## 2	Q2	4.027
## 3	Q3	4.036
## 4	Q4	4.023
## 5	Q5	4.029
## 6	Q6	4.038
## 7	Q7	4.043
## 8	Q8	4.045

After linking defaulted obligors with the characteristics of the performing population, each metric can be calculated.

The table below presents a sample of the resulting metrics.

##	quarter	default.id	rating	baseline	residual	residual.diff	rating.diff
##	Q1	Default_1	4	4.013	-0.013	NA	NA
##	Q2	Default_1	4	4.027	-0.027	-0.014	0
##	Q3	Default_1	5	4.036	0.964	0.991	1
##	Q4	Default_1	5	4.023	0.977	0.013	0
##	Q5	Default_1	5	4.029	0.971	-0.006	0
##	Q6	Default_1	5	4.038	0.962	-0.009	0

Simulation Results cont.

Averaging each metric from the previous slide, practitioners obtain the tendency-based metrics for the observed defaults.

```
##           metric      value
## 1 average_residual 1.8182500
## 2  residual_trend 0.3668571
## 3   rating_trend 0.3714286
```

Finally, the next slide presents the simulated distribution of each metric, which can be compared to zero or any other selected threshold used for evaluation.

Simulation Results cont.

Bootstrapped Distribution of Tendency Metrics

